

EE/CS 148B - Homework 3

Aiden Di Carlo

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Vision Transformer

(a) `vit_pooling`: For tasks requiring spatial reasoning, attention pooling is expected to perform best because it can learn to weight tokens selectively based on task-relevant spatial cues, rather than treating all positions equally (mean pooling) or relying on a single generic summary (CLS). Mean pooling discards per-patch identity by averaging, which collapses distinctions between, say, a red cube on the left versus the right; the CLS token suffers the same problem more acutely, since it is trained to summarize global semantics and does not preserve fine-grained spatial layout at all. When an entire image is condensed into a single CLS vector before being passed to a language model, all explicit positional relationships between objects are lost. The model receives no direct signal about where each object is, only a holistic blend of what is in the image.

(b) `vit_patch_size`:

(1) Patch counts and attention cost. For a 224×224 image the number of patches is $N = (224/P)^2$:

P	N
8	784
16	196
32	49

Self-attention scales as $O(N^2 d_{\text{model}})$, and $N \propto P^{-2}$. Thus, the overall cost scales as $O(P^{-4} d_{\text{model}})$. Halving the patch size increases attention compute by $2^4 = 16\times$, so shrinking P is extremely expensive.

(2) Wall-clock times (batch of 16 images, $d_{\text{model}} = 384$, 6 heads, 6 blocks; RTX 4070 Laptop GPU; 5 warmup + 20 timed steps):

P	Mean (ms)	Std (ms)
8	84.9	1.8
16	14.8	1.3
32	10.1	1.3

The empirical ratios ($8.4\times$ between $P = 8$ and $P = 16$, and $1.5\times$ between $P = 16$ and $P = 32$) are smaller than the theoretical $16\times$ and $256\times$ because the GPU is compute-bound only for the largest sequence length; at $P = 16$ and $P = 32$ the bottleneck shifts to memory bandwidth and kernel launch overhead, compressing the observed differences. The $P = 8$ overhead, roughly $8\times$ slower than $P = 16$, confirms that the quadratic attention term dominates at high patch counts.

(3) Accept smaller patches when fine-grained spatial fidelity is critical and the compute budget allows it (e.g., OCR, dense prediction, or counting small objects where coarser tokens would merge distinct features).

CLIP-Style Contrastive Pretraining

(a) infonce: The loss is averaged in both directions because each (image, text) pair should serve as a positive for both retrieval directions—finding the right caption given an image, and the right image given a caption. Using only one direction would bias the model toward one retrieval mode and leave half the pairwise supervision signal in each batch unused.

(b) clip_train:

Validation accuracy rises steeply during the first eight epochs ($53\% \rightarrow 87\%$), then continues to improve at a diminishing rate before plateauing near 91% from epoch 17 onward. Training loss, by contrast, decreases monotonically throughout all 20 epochs—including the final epochs where accuracy is essentially flat—indicating the model keeps tightening image-text alignment on training pairs past the point of diminishing returns for zero-shot classification. This is expected under EuroSAT’s class-template captions: many examples per batch share the same caption, so the loss can still improve by better separating near-duplicate positives even after the coarser class-level decision boundaries are learned.

(c) clip_zeroshot:

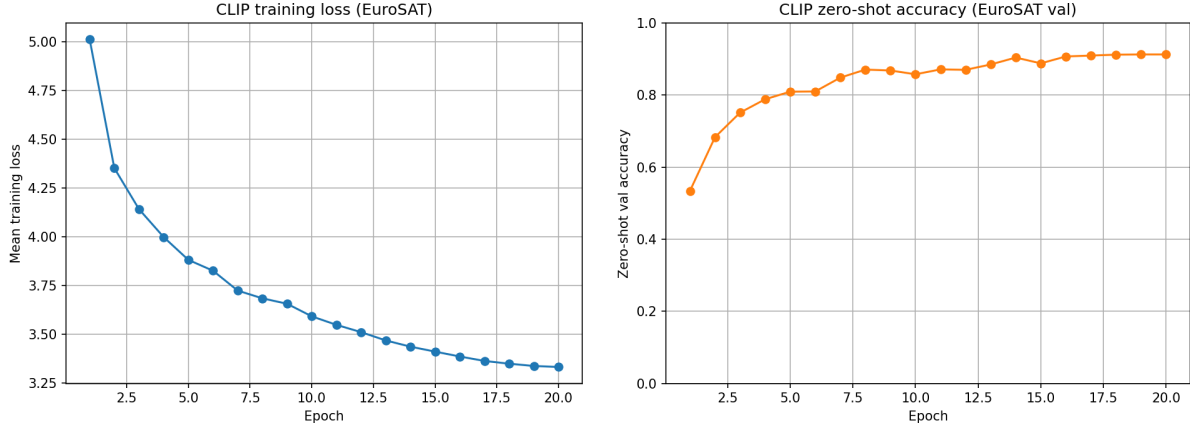


Figure 1: Left: mean training loss per epoch. Right: zero-shot validation accuracy per epoch. EuroSAT, 20 epochs, A100 GPU.



Nearly all errors are reasonable confusions between visually similar classes: the four agricultural/vegetation categories (Pasture, Herbaceous Vegetation, Annual Crop, Permanent Crop) appear as low-texture green patches at 64×64 resolution, making them hard to distinguish, and Industrial Buildings versus Residential Buildings are both regular grids of rooftops separated by roads. The one surprising error is Annual Crop predicted as Highway: at this resolution, the parallel furrows of a ploughed field closely resemble lane markings on a multi-lane road. Because all five wrong predictions still place the correct class in the top-3, the

mistakes reflect genuine local ambiguity rather than a failure of the embedding space: the model has learned semantically coherent clusters (“green land,” “built environment,” “water”) but struggles to separate fine-grained classes within each cluster.

LoRA Fine-Tuning

- (a) `lora.linear`: Applying LoRA (rank 8, $\alpha = 16$) to every `q_proj` and `v_proj` in the ViT and adding a 45-way classification head gives the following parameter breakdown:

	Count
Total parameters	11,013,165
Trainable parameters	275,373
of which LoRA A/B	258,048
of which head	17,325
Trainable / Total	2.50%

The LoRA count follows from 36 adapted heads (6 heads $\times$ 6 blocks), each contributing $2 \times (r \cdot d_{\text{model}} + d_{\text{head}} \cdot r) = 2 \times (8 \cdot 384 + 64 \cdot 8) = 7,168$ parameters, giving $36 \times 7,168 = 258,048$. Total parameters for LoRA exceeds the base model (10,755,117) by exactly this amount.

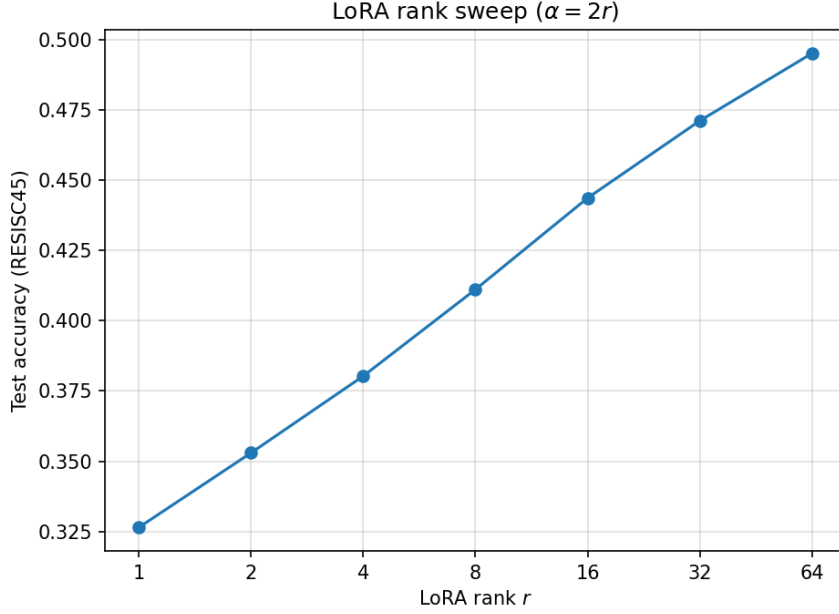
- (b) `lora.compare`:

Method	Test Acc	Trainable Params	Peak Mem (MB)	Time (s)
Linear probe	37.95%	17,325	260.7	105.4
LoRA ($r = 8$, $\alpha = 16$)	41.44%	275,373	1,210.7	139.0
Full fine-tuning	63.78%	10,755,117	1,760.5	121.6

Full fine-tuning achieves the highest accuracy (63.8%) by a large margin because EuroSAT and RESISC45 differ substantially in class structure (10 vs. 45 classes, multispectral satellite vs. aerial RGB imagery), so the frozen EuroSAT representation transfers imperfectly and requires intermediate features—not just the final head—to be updated. LoRA outperforms the linear probe by 3.5 percentage points despite training only $16\times$ more parameters (275K vs. 17K), showing that adapting the attention Q/V projections yields a more flexible adjustment of the representation than a frozen linear readout. Peak memory follows the amount of stored activations during backpropagation: linear probe requires no gradient to pass through the ViT (261 MB), while LoRA and full fine-tuning both backpropagate through every block (1,211 MB and 1,761 MB), with full fine-tuning’s extra overhead coming from larger optimizer

states for its 10M trainable parameters. Training times are similar across methods (105–139 s) because all three run the same forward pass through the ViT; the LoRA backward is nearly as expensive as full fine-tuning since gradients must still flow through all blocks to reach the A and B matrices.

(c) lora_rank:



Accuracy increases monotonically across all seven ranks (32.7%, 35.3%, 38.0%, 41.1%, 44.4%, 47.1%, 49.5% for $r = 1, 2, 4, 8, 16, 32, 64$), with per-doubling gains of roughly 2.4–3.3 percentage points. The curve is approximately log-linear. In large language models ($d_{\text{model}} \gg 10^3$), $r = 8$ or $r = 16$ is sufficient because the fine-tuning update is empirically very low-rank relative to the enormous weight matrices—the few most salient adaptation directions capture almost all of the needed shift. Our ViT has $d_{\text{model}} = 384$ and is adapting across a substantial domain gap (10-class EuroSAT satellite imagery $\rightarrow$ 45-class RESISC45 aerial imagery), so the effective rank of the update is higher relative to d_{model} : the model must adjust a broader subspace of its attention projections to separate 45 visually diverse classes, and even $r = 64$ (one-sixth of d_{model}) has not exhausted the useful adaptation directions.